From Poincaré Inequality to a Transport-Entropy Inequality

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Poincaré Inequality and Exponential Concentration

The Poincaré Inequality is a classical inequality in the study of the concentration of measure phenomenon. Given a probability measure $\mu \in \mathcal{P}\left(\mathbf{R}^d\right)$, we say that it satisfies a Poincaré Inequality with constant $C \geq 0$ (or $\mu \in \mathsf{PI}\left(C\right)$) if for all locally-Lipschitz f, there holds the bound

$$\mathbf{var}_\mu\left[f\right] \leq C \cdot \mathbf{E}_\mu\left[\left\|\nabla f\right\|^2\right].$$

One immediately sees that all uniformly-Lipschitz functions have finite variance under μ, implying at least some polynomial lightness of tails for μ.

Actually, one even has exponentially-light tails, after the following classical argument: let g be 1-Lipschitz and centered under μ. For $\theta \geq 0$, define the moment-generating function $M\left(\theta\right) = \mathbf{E}_\mu\left[\exp\left(\theta \cdot g\right)\right]$ and the cumulant-generating function $K\left(\theta\right) = \log M\left(\theta\right)$. Taking $f = \exp\left(\frac{\theta}{2} \cdot g\right)$ in the Poincaré Inequality, compute that

$$\begin{aligned}
M\left(\theta\right) - M\left(\frac{\theta}{2}\right)^2 &\leq \frac{C}{4} \cdot \theta^2 \cdot M\left(\theta\right) \\
\implies \qquad M\left(\theta\right) &\leq \frac{1}{1 - \frac{C}{4} \cdot \theta^2} \cdot M\left(\frac{\theta}{2}\right)^2 \\
\implies \qquad K\left(\theta\right) &\leq 2 \cdot K\left(\frac{\theta}{2}\right) + \log\left(\frac{1}{1 - \frac{C}{4} \cdot \theta^2}\right),
\end{aligned}$$

i.e. the cumulant-generating function of $g\left(X\right)$, $X \sim \mu$ has a sort of self-bounding structure. Applying this relation recursively, one computes that for any natural number P, there holds the refined bound

$$K\left(\theta\right) \leq 2^P \cdot K\left(\frac{\theta}{2^P}\right) + \sum_{0 \leq p < P} 2^p \cdot \log\left(\frac{1}{1 - \frac{C}{4} \cdot \left(\frac{\theta}{2^p}\right)^2}\right).$$

Recalling that $K\left(\theta\right) = \frac{\mathbf{var}_\mu[g]}{2} \cdot \theta^2 \cdot \left(1 + \mathcal{O}\left(\theta\right)\right)$ as $\theta \to 0^+$, one argues that for a fixed θ,

$$0 \leq 2^P \cdot K\left(\frac{\theta}{2^P}\right) = 2^{-P} \cdot \left\{\frac{\mathbf{var}_\mu\left[g\right]}{2^1} \cdot \theta^2 \cdot \left(1 + \mathcal{O}\left(2^{-P}\right)\right)\right\} \in o\left(1\right) \qquad \text{as } P \to \infty.$$

Sending $P \to \infty$, one can thus neglect the first term and obtain an upper bound in the form of an infinite sum, i.e.

$$K\left(\theta\right) \leq \sum_{p \geq 0} 2^p \cdot \log\left(\frac{1}{1 - \frac{C}{4} \cdot \left(\frac{\theta}{2^p}\right)^2}\right).$$

This upper bound is somewhat unwieldy at a first glance. Using that $\vartheta \mapsto \vartheta^{-1} \cdot \log\left(\frac{1}{1-\vartheta}\right)$ is non-decreasing on $\mathbf{R}_+$, one can exhibit the term-wise bound

$$p \geq 0 \implies \log\left(\frac{1}{1 - \frac{C}{4} \cdot \left(\frac{\theta}{2^p}\right)^2}\right) \leq 2^{-2 \cdot p} \cdot \log\left(\frac{1}{1 - \frac{C}{4} \cdot \theta^2}\right),$$

from which a simple geometric sum implies that

$$K(\theta) \leq 2 \cdot \log\left(\frac{1}{1 - \frac{C}{4} \cdot \theta^2}\right)$$

As such, it holds that whenever $\|h\|_{\mathrm{Lip}} \leq 1$, $|\theta| < \frac{2}{\sqrt{C}}$,

$$\log\left(\mathbf{E}_\mu\left[\exp\left(\theta \cdot \{h - \mathbf{E}_\mu[h]\}\right)\right]\right) \leq 2 \cdot \log\left(\frac{1}{1 - \frac{C}{4} \cdot \theta^2}\right).$$

In particular, taking $h(x) = \|x\|_2$, one sees that $\mathbf{P}_\mu\left(\|X\|_2 \geq \mathbf{E}\left[\|X\|_2\right] + t\right) \in \exp\left(-\Omega(t)\right)$.

Transport Inequalities

It is well-known since at least a work of Bobkov and Götze (and subsequently developed ably by Gozlan and Léonard) that exponential concentration of Lipschitz functions is equivalent to a certain functional inequality known as a 'transport-entropy inequality'. Such inequalities take the form

$$\forall \nu \in \mathcal{P}\left(\mathbf{R}^d\right), \qquad \alpha\left(\mathcal{T}_1(\nu, \mu)\right) \leq \mathsf{KL}(\nu, \mu),$$

where $\alpha : \mathbf{R}_+ \to \mathbf{R}_+$ is an increasing, convex function satisfying $\alpha(0) = 0$, $\mathcal{T}_1$ denotes the Kantorovich transport distance, and KL denotes the relative entropy or 'Kullback-Leibler divergence'. We will show how the Poincaré inequality implies such an inequality, directly carrying on from where the previous proof left off, and following a strategy which I first learned of from lecture notes of Van Handel.

To this end, recall that we have established that (writing $K_C(\theta) = 2 \cdot \log\left(\frac{1}{1 - \frac{C}{4} \cdot \theta^2}\right)$)

$$\sup_{\|h\|_{\mathrm{Lip}} \leq 1} \sup_{|\theta| < \frac{2}{\sqrt{C}}} \left\{\log\left(\mathbf{E}_\mu\left[\exp\left(\theta \cdot \{h - \mathbf{E}_\mu[h]\}\right)\right]\right) - K_C(\theta)\right\} \leq 0.$$

The classical Donsker-Varadhan principle grants that for any sufficiently regular ξ, there holds the equality

$$\log\left(\mathbf{E}_\mu[\exp \xi]\right) = \sup\left\{\mathbf{E}_\nu[\xi] - \mathsf{KL}(\nu, \mu) : \nu \in \mathcal{P}\left(\mathbf{R}^d\right)\right\},$$

and so our initial bound on the cumulant-generating function can be written as

$$\sup_{\nu \in \mathcal{P}(\mathbf{R}^d)} \sup_{\|h\|_{\mathrm{Lip}} \leq 1} \sup_{|\theta| < \frac{2}{\sqrt{C}}} \left\{\theta \cdot \{\mathbf{E}_\nu[h] - \mathbf{E}_\mu[h]\} - \mathsf{KL}(\nu, \mu) - K_C(\theta)\right\} \leq 0.$$

Recalling also that

$$\sup_{\|h\|_{\mathrm{Lip}} \leq 1} \left|\mathbf{E}_\nu[h] - \mathbf{E}_\mu[h]\right| = \mathcal{T}_1(\nu, \mu),$$

we may optimise over h to say that equivalently,

$$\sup_{\nu \in \mathcal{P}(\mathbf{R}^d)} \sup_{|\theta| \in \left[0, \frac{2}{\sqrt{C}}\right)} \left\{|\theta| \cdot \mathcal{T}_1(\nu, \mu) - \mathsf{KL}(\nu, \mu) - K_C(\theta)\right\} \leq 0.$$

Optimising over θ requires some additional algebra. Check first that in the case of $C = 4$, for $\eta \in \mathbf{R}$, it holds that

$$\begin{aligned}\sup\left\{\vartheta \cdot \eta - \frac{1}{2} \cdot K_4(\vartheta) : |\vartheta| < 1\right\} &= \sup\left\{\vartheta \cdot \eta - \log\left(\frac{1}{1 - \vartheta^2}\right) : |\vartheta| < 1\right\} \\ &= \sqrt{1 + \eta^2} - 1 - \log\left(\frac{1 + \sqrt{1 + \eta^2}}{2}\right) =: \frac{1}{2} \cdot \alpha_4(\eta).\end{aligned}$$

Given this, we can handle general $C > 0$ by writing that

$$\begin{aligned}\sup\left\{\theta\cdot\eta - K_C(\theta) : |\theta| < \frac{2}{\sqrt{C}}\right\} &= 2\cdot\sup\left\{\frac{\theta\cdot\sqrt{C}}{2}\cdot\frac{\eta}{\sqrt{C}} - \frac{1}{2}\cdot K_C(\theta) : \frac{\theta\cdot\sqrt{C}}{2} < 1\right\} \\ &= 2\cdot\sup\left\{\vartheta\cdot\frac{\eta}{\sqrt{C}} - \frac{1}{2}\cdot K_4(\vartheta) : |\vartheta| < 1\right\} \\ &= 2\cdot\left\{\frac{1}{2}\cdot\alpha_4\left(\frac{\eta}{\sqrt{C}}\right)\right\} =: \alpha_C(\eta).\end{aligned}$$

As such, optimising over θ yields the the equivalent form

$$\sup_{\nu\in\mathcal{P}(\mathbf{R}^d)}\left\{\alpha_C\left(\mathcal{T}_1(\nu,\mu)\right) - \mathsf{KL}(\nu,\mu)\right\} \le 0,$$

i.e. for all $\nu \in \mathcal{P}\left(\mathbf{R}^d\right)$, with $\eta = \frac{\mathcal{T}_1(\nu,\mu)}{\sqrt{C}}$, it holds that

$$2\cdot\left\{\sqrt{1+\eta^2} - 1 - \log\left(\frac{1+\sqrt{1+\eta^2}}{2}\right)\right\} = \alpha_C\left(\mathcal{T}_1(\nu,\mu)\right) \le \mathsf{KL}(\nu,\mu).$$

A Simplified Inequality

The specific form of α arising from this proof is not the easiest to work with (e.g. it is not easily invertible without recourse to special functions). In this section, we study a simplified lower bound with various favourable properties

$$\alpha_4(\eta) \ge \underline{\alpha}_4(\eta) := 2\cdot\left\{\sqrt{1+\frac{1}{2}\cdot\eta^2} - 1\right\}.$$

We will later see that this is extremely accurate for small η, and is off by a multiplicative factor of $\asymp \sqrt{2}$ for large η. Moreover, it is elementary to invert; check that

$$\alpha_4^{-1}(z) = \sqrt{2\cdot z + \frac{1}{2}\cdot z^2} \le \sqrt{\frac{2\cdot z}{1-\frac{1}{4}\cdot z}}.$$

In view of this bound, we can write

$$2\cdot\left\{\sqrt{1+\frac{1}{2\cdot C}\cdot\mathcal{T}_1(\nu,\mu)^2} - 1\right\} = \underline{\alpha}_4\left(\frac{\mathcal{T}_1(\nu,\mu)}{\sqrt{C}}\right) \le \mathsf{KL}(\nu,\mu),$$

and hence given an upper bound on $\mathsf{KL}(\nu,\mu)$, it is straightforward to control $\mathcal{T}_1(\nu,\mu)$ as

$$\mathcal{T}_1(\nu,\mu) \le \sqrt{C\cdot\left\{2\cdot\mathsf{KL}(\nu,\mu) + \frac{1}{2}\cdot\mathsf{KL}(\nu,\mu)^2\right\}} \le \sqrt{2\cdot C\cdot\frac{\mathsf{KL}(\nu,\mu)}{1-\frac{\mathsf{KL}(\nu,\mu)}{4}}},$$

which is often how these inequalities are used in applications.

Proof of the Lower Bound

To prove this bound, we begin by noting that

$$\alpha_4''(\eta) = \frac{2}{\sqrt{1+\eta^2}\cdot\left\{1+\sqrt{1+\eta^2}\right\}}, \quad \underline{\alpha}_4''(\eta) = \frac{1}{\left(1+\frac{1}{2}\cdot\eta^2\right)^{\frac{3}{2}}},$$

and so if we can order these two derivatives, then integrating up twice from $\eta = 0$ will yield the desired result (since α_4 and $\underline{\alpha}_4$ are readily seen to match well for small η).

To this end, we start with some elementary algebraic observations: by the Cauchy-Schwarz inequality, for any real ξ, one sees readily that

$$1+\xi^6 \geq 2\cdot\xi^3, \qquad \xi^2+\xi^4 \geq 2\cdot\xi^3.$$

Adding up these two inequalities and re-arranging allows one to infer that

$$\left(1+\xi^2\right)^3 \geq 2\cdot\xi^2\cdot(1+\xi)^2 \implies 2\cdot\left(\frac{1+\xi^2}{2}\right)^{\frac{3}{2}} \geq \xi\cdot(1+\xi)\,.$$

Setting $\xi^2 = 1+\eta^2$ and re-arranging further then yields that

$$2\cdot\left(1+\frac{1}{2}\cdot\eta^2\right)^{\frac{3}{2}} \geq \sqrt{1+\eta^2}\cdot\left(1+\sqrt{1+\eta^2}\right)$$

which is tantamount to our claim.

Observe that the approximation is really quite accurate; for $\eta\sim 0$, one has that

$$\alpha_4''(\eta) \sim 1-\frac{3}{4}\cdot\eta^2+\frac{5}{8}\cdot\eta^4, \qquad \underline{\alpha}_4''(\eta) \sim 1-\frac{3}{4}\cdot\eta^2+\frac{15}{32}\cdot\eta^4,$$

so that $\alpha_4''(\eta)-\underline{\alpha}_4''(\eta)\sim\frac{5}{32}\cdot\eta^4$, and hence that $\alpha_4(\eta)-\underline{\alpha}_4(\eta)\sim\frac{1}{192}\cdot\eta^6$.